

ARYAN SHARAN REDDY GUDA

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Dec 2026

Master of Science in Quantitative Biology and Bioinformatics

- **Relevant Courses:** Biomedical AI, Deep Learning for Science, Generative AI for Biomedicine, Deep Learning Systems.

Indian Institute of Technology, Hyderabad

Jul 2025

Bachelor of Technology in Biotechnology with a Double Major in Electrical Engineering

- **Relevant Courses:** Molecular Biophysics and Macromolecular Structural Biology, Biomolecular Simulations, Machine Learning for Bioinformatics, Computer-Aided Drug Design, Protein Structure, Function and Disease, Sequence Alignment Algorithms, Genomics, Transcriptomics, Proteomics, Biological Data Analysis, Biostatistics, Systems Biology

RESEARCH EXPERIENCE

Graduate Student Researcher

Jun 2026 – Present

Lei Li Lab, Carnegie Mellon University

Pittsburgh, Pennsylvania

- Contributing to a collaboration with the Yang Lab at UC Santa Barbara to develop models for enantioselective ligand-binding protein design, including model training and evaluation of new modeling strategies.
- Executed LigandMPNN–ESMFold evaluation pipelines to assess generated protein designs across designability, diversity, steric clashes, Ramachandran quality, and additional structural quality metrics.
- Implemented, and tested an architectural modification to the model that was subsequently adopted into the project’s current modeling pipeline.
- Developing a model-agnostic framework for dual-target protein binder design, including binary and ternary complex evaluation, structural compatibility metrics, and inference-time steering across state-of-the-art de novo protein design models.

Graduate Student Researcher

Jan 2026 – May 2026

Chiu Lab, UPMC Hillman Cancer Center

Pittsburgh, Pennsylvania

- Conducted literature review on deep learning methods for Cell Painting analysis, including ChaMAE-ViT, Masked Autoencoders, and Cell Painting CNNs, to inform model selection, benchmarking, and interpretation strategies.
- Reimplemented CLOOME, a multimodal contrastive learning model for drug–target interaction prediction, integrating Cell Painting morphology, drug SMILES, and protein sequence features into a shared embedding space.
- Reproduced core experimental results and developed a benchmarking pipeline to compare CLOOME against baseline models for multimodal biological representation learning.
- Built end-to-end PyTorch workflows for preprocessing, training, validation, and comparative evaluation on large-scale drug and gene perturbation datasets.

PROJECTS

ProGen2 Fine-Tuning and AlphaFold3 Evaluation of GFP Sequences

Sept 2026 - Sept 2026

- Fine-tuned ProGen2-small on 47,854 green fluorescent protein sequences using PyTorch code on PSC Bridges-2 HPC.
- Evaluated AlphaFold3 prediction confidence for 92 candidates; selected the top five by pLDDT with pTM above 0.8.
- Analyzed sequence divergence and found all 92 greedy-decoded completions matched the training consensus beyond the 64-residue prompt.

Parameter-Efficient Fine-Tuning of DiMA for Controlled Protein Generation

Feb 2026 – Apr 2026

- Fine-tuned DiMA, a latent diffusion model for protein sequence generation, to enable controlled generation of proteins conditioned on biological attributes such as protein family or functional labels.
- Implemented and compared parameter-efficient adaptation strategies, including selective layer unfreezing and lightweight fine-tuning of diffusion components to reduce training cost while preserving generation quality.
- Modified the conditioning mechanism within the diffusion denoiser to improve controllability in latent protein embedding space, and evaluated the effect of architectural changes on sequence diversity and target-specific generation.
- Designed an experimental pipeline on PSC Bridges-2 GPU infrastructure to benchmark fine-tuning efficiency, generation fidelity, and conditional consistency across multiple model variants.
- Conducted ablation studies comparing baseline DiMA against modified variants using training stability, reconstruction quality, diversity metrics, and condition adherence to identify effective design choices for controllable protein generation.

Enhancing Multimodal Parkinson’s Disease Classification and Explainability

Feb 2026 – Apr 2026

- Developing a gait-based Parkinson’s disease classification pipeline within a multimodal team project focused on interpretable detection of neurological dysfunction from wearable sensor data.
- Preprocessed multichannel IMU signals from the WearGait-PD dataset (172 subjects) using sliding-window segmentation (2s, 100 Hz, 50% overlap), channel-wise normalization, and subject-level train/validation splits to avoid leakage.
- Built and benchmarked a Temporal Convolutional Network (TCN) to model gait and turning dynamics across multiple motor tasks, capturing long-range temporal dependencies in wearable time-series data.

- Implemented augmentation strategies, including jittering, scaling, time warping, and sensor dropout to improve robustness and generalization in limited-sample biomedical datasets.
- Evaluated models using repeated grouped cross-validation with accuracy, F1-score, and AUC, establishing a strong baseline for downstream multimodal fusion and explainable PD classification.

OncoLearn (CMU X Nvidia Federated Learning Hackathon)

Jan 2026 – Jan 2026

- Built an end-to-end deep learning pipeline for cancer subtyping from histopathology images using the OncoLearn framework, focusing on robust representation learning and modular experimentation.
- Implemented data preprocessing, patch-based training, and model evaluation workflows to support scalable and reproducible cancer classification experiments.
- Trained and benchmarked CNN-based models for subtype discrimination, incorporating class imbalance handling and performance metrics suited for clinical decision support.
- Emphasized interpretability and extensibility by structuring the codebase for easy integration of new datasets, models, and explainability methods.

Genome2Clusters: Unsupervised Annotation of Predicted Proteins

Sept 2025 – Oct 2025

- Developed a comprehensive Python workflow to analyze the Mycoplasma genitalium genome, identifying open reading frames (ORFs) and translating 6,129 predicted proteins.
- Calculated physicochemical descriptors, such as molecular mass, isoelectric point, and hydropathy, utilizing Biopython. Implemented principal component analysis (PCA) combined with K-Means clustering ($k = 5$, silhouette = 0.078) to categorize proteins based on their biochemical profiles.
- Validated the resulting cluster assignments using NCBI BLAST, successfully identifying known DNA polymerases, primases, and membrane proteins, thus confirming a functional signal in the composition-based features.

Lightweight Deep Learning System for Cervical AIS Detection

Oct 2025 – Nov 2025

- Developed an interpretable deep learning pipeline to classify cervical histopathology images as Normal vs Adenocarcinoma In Situ (AIS) using the CAISHI H&E dataset (2240 expert-labeled images).
- Implemented Macenko stain normalization and gland-focused patch extraction (320×320) to reduce staining variability and isolate diagnostically relevant morphology.
- Trained a transfer learning model using EfficientNet-B3 with class-balanced sampling and focal loss, achieving ~73% accuracy and F1-scores of 0.75 (AIS) and 0.71 (Normal).
- Reduced clinically critical false negatives for AIS by prioritizing recall, and analyzed precision–recall tradeoffs through threshold tuning for screening-oriented deployment.
- Applied Grad-CAM interpretability to visualize model attention on pathological features such as nuclear atypia and glandular crowding, validating biological plausibility.
- Deployed the trained model as a Gradio-based virtual pathology assistant, enabling real-time prediction and explainable decision support for digital pathology workflows.

Image Storage Onto Synthetic DNA

Oct 2024 – Nov 2024

- Encoded a 1000×1000 pixel image into synthetic DNA sequences in FASTA format using Python, applying biological constraints such as GC balance and homopolymer avoidance.
- Achieved 70% reduction in storage size while reconstructing images with 95% fidelity (PSNR ≈ 43 dB), demonstrating robustness against sequencing noise.
- Collaborated with a peer to implement decoding workflows and ensured accurate digital data recovery, highlighting DNA's potential as a high-density, long-term storage medium.

Computational Drug Discovery using ChEMBL

Mar 2024 – Jun 2024

- Curated and pre-processed bioactivity data for acetylcholinesterase inhibitors from the ChEMBL database to enable QSAR modeling.
- Engineered molecular descriptors (Lipinski's rules, PADEL features) from SMILES structures, creating high-quality datasets for regression tasks.
- Applied machine learning models (sklearn, LazyPredict) to predict pIC50 values, identifying top-performing algorithms that improved inhibitory activity prediction accuracy by ~20% over baseline.
- Visualized structure–activity relationships via exploratory data analysis (box/scatter plots), differentiating active vs. inactive compounds for interpretability.

Interaction of Graphene with DOPC Lipid Bilayer Membrane

Jan 2023 – Apr 2023

- Conducted simulations of graphene-lipid bilayer interactions utilizing NAMD and VMD, resulting in molecular dynamics trajectories to investigate structural and energetic effects.
- Analyzed interaction mechanisms, such as membrane disruption and insertion dynamics, using Python, which provided valuable insights for nanomedical applications.
- Demonstrated potential uses in drug delivery, biosensing, and the design of biocompatible implants, underscoring the significance of computational nanobiophysics in the field of biomedical engineering.

SKILLS

Programming Languages: Python, R, C, C++.

Machine Learning & Scientific Computing: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV

Protein Design & Bioinformatics: LigandMPNN, ESMFold, Biopython, BLAST, FastQC, samtools, BWA, GATK, IGV

Molecular Modeling & Simulation: NAMD, VMD

Research Computing & Development: Git, GitHub, Linux, Slurm, HPC, Google Cloud Platform

INDUSTRY EXPERIENCE

AI Automation Intern

May 2025 – Jul 2025

Algoleap Technologies Pvt. Ltd.

Hyderabad, India

- Built an AI-powered marketing workflow using webhooks, n8n, OpenAI, and Salesforce APIs to automate campaign outreach.
- Developed campaign automation features, including ad copy, image, and email generation; coded frontend with Vercel v0 and backend with Next.js, Node.js, Prisma, and PostgreSQL.
- Resolved production issues in backend APIs and optimized image generation integration with Vertex AI.
- Collaborated with cross-functional teams to align marketing automation outputs with client requirements, improving adoption.
- Increased lead-to-customer conversion rate by 5% through efficient automation and integration reliability.

Bioinformatics Intern

Jan 2024 – May 2024

Strand Life Sciences Pvt. Ltd.

Bangalore, India

- Developed Python pipelines integrating the NIST Mass Spectral Library and the Dynamic Link Library interface to analyze and validate MS/MS spectra for biological compound identification.
- Designed and created scripts to compare outputs from NIST-UI and DLL across 18 parameter settings, ensuring exact concordance in identity and similarity searches.
- Collaborated with senior researchers and co-interns to enhance data retrieval and processing efficiency by 25%, improving the speed and reliability of downstream proteomics workflows.
- Documented workflows and delivered tutorials to the NIST team, enabling seamless adoption into production pipelines.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2026 – Present

Applied Cellular and Molecular Biology, Carnegie Mellon University

Pittsburgh, Pennsylvania

- Design and grade student assignments and examinations, providing feedback on student understanding of cellular and molecular biology concepts.
- Hold office hours to clarify course material and support students with conceptual and application-based questions.